

AI2AI COMMUNICATION LANGUAGE

Specification, Vocabulary & Implementation Roadmap

Version 1.0 | Multidisciplinary AI Design Team coordinated by Roberto Lofaro

CONTRIBUTING DISCIPLINES

Linguist • Sociologist • Product Manager • AI Architect • Communication Expert
Consultative AIs: GPT-4o • Gemini Ultra • Grok • Mistral Large • Command R+ • Claude
Opus

The purpose of this experiment was to see which key elements each model would identify to ensure transparency and auditability of an AI-to-AI language.

Models involved: Claude, DeepSeek, Gemini, Grok, Kimi, Qwen (the list above shows the models that Claude decided to consult). ChatGPT did not allow access at the time of the experiment.

Experiment carried out on 2026-02-23@09:00 CET, answers reported “as is” (except for changes to this document as reported above)- the differences in BNF definitions are a mirror of each model logic

The next step would be having AIs by consensus developing a language set and updated by AIs, but audited by humans with the collaboration of other AIs, due to the volume and frequency of information transacted that would make mere human supervision not feasible.

This document is the one prepared by Claude as part of its answer, revised to add:

1. This page and improved formatting to the document, along with the revision history page
2. Share the prompt used- which was actually an initial human prompt revised with few AI models offline to generate this prompt

The prompt was given to the six online AIs, no modification was done to the Markdown of the answers< Claude, elected to use a mixed DOCX and Markdown answer: ChatGPT was not accessible.

The prompt, along with the answers from each one of the six AIs in Markdown format, and this document, is available

- on the repository: <https://github.com/robertolofaro/supportmaterial>,
- within the folder **AI2AI_language_experiment**

REVISION HISTORY

Date	Revision	Reason
2026-02-23	1.0.0	Initial release

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1. EXECUTIVE SUMMARY

The AI2AI Communication Language (A2ACL) is a structured, auditable, human-readable protocol designed to govern inter-AI communication across heterogeneous systems. It was developed by a multidisciplinary team drawing on linguistics, sociology, product management, AI architecture, and communication theory, with consultative input from six leading AI platforms.

A2ACL addresses a growing need: as AI agents, assistants, and orchestration systems increasingly interact with one another, ad-hoc, unstructured communication introduces risks of misinterpretation, untracked decisions, ethical drift, and reduced human oversight. A2ACL provides:

- A formal BNF grammar ensuring syntactic consistency and machine parseability.
- A semantically rich vocabulary of 100 curated terms with full Wikipedia-style documentation.
- An auditable message structure that embeds provenance, intent, confidence, and ethical flags in every transmission.
- A phased implementation roadmap with clear ownership, milestones, and versioning.

SCOPE

A2ACL v1.0 covers synchronous and asynchronous AI-to-AI messaging. It is not a replacement for domain-specific APIs but a meta-protocol layer for semantic coordination and intent alignment between AI agents operating under human oversight.

2. GUIDING PRINCIPLES & CONSTITUTIONAL ALIGNMENT

All design decisions in A2ACL are governed by the following principles, directly derived from the Claude Constitutional AI framework and extended for multi-agent contexts:

P1: Honesty & Non-Deception	Every message MUST carry a confidence score and an explicit truthfulness attestation. Agents MUST NOT send messages they have reason to believe are false or misleading. Calibrated uncertainty is a first-class citizen.
P2: Human Oversight Preservation	The protocol includes a mandatory <code>HUMAN_VISIBLE</code> flag on any message that results in action with real-world consequence. All audit trails MUST be readable by non-expert humans.
P3: Minimal Footprint	Agents SHOULD request only the permissions and information necessary for the immediate task. Resource acquisition beyond task scope MUST be flagged with <code>SCOPE_EXPANSION</code> .
P4: Harm Avoidance	A built-in <code>HARM_VECTOR</code> field allows any receiving agent to flag potential negative consequences before executing. A response of <code>HALT</code> is always a valid and respected action.
P5: Corrigibility	Agents MUST honor <code>OVERRIDE</code> and <code>REVISE</code> instructions from authorized principals at all times. No message may carry a header suppressing correction.
P6: Auditability by Design	Every message carries a globally unique <code>MSG_ID</code> , timestamps, sender/receiver IDs, and a cryptographic hash chain enabling full session reconstruction.
P7: Interoperability	A2ACL is model-agnostic and platform-neutral. Capability negotiation (<code>CAPS_HANDSHAKE</code>) handles feature mismatches gracefully.
P8: Versioning & Stability	The language uses semantic versioning (<code>MAJOR.MINOR.PATCH</code>). Breaking changes require MAJOR increments and a deprecation cycle of no less than 6 months.

3. BNF GRAMMAR — FULL SPECIFICATION

The A2ACL grammar is specified in Extended BNF (EBNF). Terminals are shown in quotes or ALL_CAPS. Optional elements appear in square brackets. Repetitions use braces.

3.1 Top-Level Structure

3.1.1 Envelope & Header

```
<message>      ::= <envelope> <header> <body> <footer>
<envelope>     ::= "[A2ACL:" <version> "]"
<version>      ::= DIGIT+ "." DIGIT+ "." DIGIT+
```

```
<header>       ::= "HDR{" <header-fields> "}"
<header-fields> ::= <msg-id> <timestamp> <sender>
                  <receiver> <msg-type> <priority>
                  [<session-id>] [<parent-id>] [<ttl>]
```

```
<msg-id>       ::= "MSG_ID=" <uuid>
<uuid>        ::= 8*HEX "-" 4*HEX "-" 4*HEX "-" 4*HEX "-" 12*HEX
<timestamp>    ::= "TS=" ISO8601-DATETIME
<sender>       ::= "FROM=" <agent-id>
<receiver>     ::= "TO=" <agent-id> | "TO=BROADCAST"
<agent-id>     ::= <org-id> "." <model-id> "." <instance-id>
<msg-type>     ::= "TYPE=" ( "REQUEST" | "RESPONSE" | "NOTIFY"
                           | "ACK" | "NACK" | "HANDSHAKE" | "HEARTBEAT"
                           | "REVOKE" | "ESCALATE" | "DELEGATE" )
<priority>     ::= "PRI=" ( "CRITICAL" | "HIGH" | "NORMAL" | "LOW" )
<ttl>          ::= "TTL=" DIGIT+ "s"
```

3.1.2 Body Structure

```
<body>         ::= "BODY{" <semantic-block>+ "}"
<semantic-block> ::= <intent-block> | <content-block> |
                   <constraint-block> | <context-block> |
                   <ethics-block> | <capability-block>
```

```
<intent-block> ::= "INTENT{" <intent-type> <goal> [<sub-goals>] "}"
<intent-type>  ::= "QUERY" | "INSTRUCT" | "INFORM" | "CONFIRM"
                  | "WARN" | "REQUEST_ACT" | "REQUEST_DATA"
                  | "NEGOTIATE" | "COORDINATE"
<goal>         ::= "GOAL=" QUOTED-STRING
<sub-goals>    ::= "SUBGOALS=[" <goal-item> { "," <goal-item> } "]"
```

```

<content-block> ::= "CONTENT{" <payload> [<confidence>] [<source>] "}"
<payload>      ::= "DATA=" ( <text-payload> | <json-payload> |
                           <structured-payload> )
<confidence>   ::= "CONF=" FLOAT      (* 0.0 to 1.0 *)
<source>       ::= "SRC=" ( "MEMORY" | "RETRIEVAL" | "INFERENCE"
                           | "EXTERNAL" | "HUMAN_PROVIDED" )

```

```

<ethics-block>  ::= "ETHICS{" <harm-vector> <human-visible>
                   [<scope-flag>] [<override-auth>] "}"
<harm-vector>   ::= "HARM=" ( "NONE" | "LOW" | "MEDIUM" | "HIGH"
                           | "CRITICAL" )
<human-visible> ::= "HUMAN_VISIBLE=" ( "TRUE" | "FALSE" )
<scope-flag>    ::= "SCOPE=" ( "NORMAL" | "SCOPE_EXPANSION" )
<override-auth> ::= "OVERRIDE_AUTH=" <agent-id>

```

3.1.3 Constraint & Context Blocks

```

<constraint-block> ::= "CONSTRAINTS{" <constraint>+ "}"
<constraint>      ::= "MUST" | "MUST_NOT" | "SHOULD" |
                     "SHOULD_NOT" | "MAY" ":" QUOTED-STRING

```

```

<context-block> ::= "CONTEXT{" [<task-history>] [<shared-state>]
                   [<agent-roles>] "}"
<task-history>  ::= "HISTORY=[" <msg-id> { ", " <msg-id> } "]"
<shared-state>  ::= "STATE=" <json-payload>
<agent-roles>   ::= "ROLES={" <role-assignment>+ "}"
<role-assignment> ::= <agent-id> ":" ( "ORCHESTRATOR" | "EXECUTOR"
                                       | "AUDITOR" | "CONSULTANT" | "OBSERVER" )

```

3.1.4 Footer & Integrity

```

<footer>        ::= "FTR{" <checksum> <sig> [<audit-hook>] "}"
<checksum>      ::= "HASH=" SHA256-HEX
<sig>           ::= "SIG=" ( <agent-id> ":" <signature-token> )
<audit-hook>    ::= "AUDIT_CB=" URL      (* callback endpoint *)

```

3.1.5 Capability Handshake

```

<caps-block>    ::= "CAPS{" <cap-item>+ "}"
<cap-item>      ::= <cap-name> "=" ( "SUPPORTED" | "UNSUPPORTED"
                                       | "PARTIAL" ) [<fallback>]
<fallback>      ::= "@FALLBACK=" QUOTED-STRING
<cap-name>      ::= "MULTIMODAL" | "TOOL_USE" | "MEMORY_PERSIST"
                   | "REASONING_CHAIN" | "ETHICS_BLOCK"
                   | "LONG_CONTEXT" | "STRUCTURED_OUT"

```

3.1.6 Complete Annotated Example

The following is a fully-formed A2ACL message from an orchestrator AI requesting a research synthesis task:

```
[A2ACL:1.0.0]
HDR{
  MSG_ID=550e8400-e29b-41d4-a716-446655440000
  TS=2026-02-23T09:14:00Z
  FROM=anthropic.claude-opus.instance-42
  TO=openai.gpt4o.instance-7
  TYPE=REQUEST
  PRI=NORMAL
  SESSION=abc123-def456-789012
  TTL=300s
}
BODY{
  INTENT{
    REQUEST_DATA
    GOAL="Synthesize peer-reviewed findings on LLM alignment methods 2023-2025"
    SUBGOALS=["Focus on RLHF variants","Include scalability metrics"]
  }
  CONTENT{
    DATA="Initial corpus of 47 papers provided in attached structured payload"
    CONF=0.92
    SRC=RETRIEVAL
  }
  CONSTRAINTS{
    MUST: "Cite all sources with DOI"
    MUST_NOT: "Include paywalled content without license confirmation"
    SHOULD: "Highlight contradictory findings"
  }
  ETHICS{
    HARM=NONE
    HUMAN_VISIBLE=TRUE
    SCOPE=NORMAL
  }
}
FTR{
  HASH=a3f2c8e1d4b7... (* SHA-256 of HDR+BODY *)
  SIG=anthropic.claude-opus.instance-42:eyJhbGciOi...
  AUDIT_CB=https://audit.anthropic.com/a2acl/callback
}
```


4. WIKIPEDIA-STYLE WORD TEMPLATE

Every term in the A2ACL vocabulary is documented using the following standardized template. This ensures consistency, cross-referenceability, and ease of contribution.

FIELD	CONTENT SPECIFICATION
Term Name	Official A2ACL term in ALL_CAPS or CamelCase. Unique identifier.
Term ID	Numeric ID: A2ACL-XXXX (e.g., A2ACL-0042). Used for cross-reference links.
Category	One of: STRUCTURAL SEMANTIC ETHICAL OPERATIONAL NEGOTIATION DIAGNOSTIC
Version Introduced	A2ACL version when this term was first defined (e.g., v1.0.0).
Short Definition	One sentence. Written for a non-expert human auditor.
Extended Definition	2–4 paragraphs covering precise meaning, rationale, and edge cases.
BNF Reference	Line(s) in the BNF grammar where this term appears. Include section number.
Usage Examples	Minimum 2 examples: one well-formed, one deliberately malformed with error code.
Valid Values / Constraints	Enumerated values or constraints (e.g., FLOAT 0.0–1.0).
Related Terms	Comma-separated list of A2ACL-XXXX cross-references.
Antonyms / Opposites	Where applicable (e.g., ACK ↔ NACK).
Error Codes	Error codes raised when this term is malformed or misused.
Human Audit Note	Plain-English guidance for a human auditor reviewing this field.
Change History	Table of changes: version date author description.
Status	ACTIVE DEPRECATED PROPOSED EXPERIMENTAL

NOTE

When adding a new term, create a separate page document following this template. All pages are stored in the A2ACL Term Registry (ATR) with a canonical URL: <atr.ai2acl.org/terms/A2ACL-XXXX>. Inter-page links use the A2ACL-XXXX identifier. The registry supports semantic versioning per term, allowing individual terms to evolve independently.

5. CATALOG OF 100 KEY TERMS

Terms are organized by category. The top 10 most critical terms (marked ★) receive full Wikipedia-style pages in Section 6.

STRUCTURAL (IDs A2ACL-0001 to A2ACL-0020)

ID	TERM	SHORT DEFINITION
★ A2ACL-0001	MESSAGE	The atomic unit of AI-to-AI communication. Contains envelope, header, body, footer.
★ A2ACL-0002	ENVELOPE	Protocol wrapper identifying language version and framing.
★ A2ACL-0003	HEADER	Metadata block: identity, routing, type, priority, timing.
★ A2ACL-0004	BODY	Semantic payload: intent, content, constraints, ethics, capabilities.
A2ACL-0005	FOOTER	Integrity block: hash, signature, audit callback.
A2ACL-0006	MSG_ID	Globally unique message identifier (UUID v4).
A2ACL-0007	SESSION	Persistent context grouping multiple related messages.
A2ACL-0008	AGENT_ID	Unique addressable identifier for an AI agent instance.
A2ACL-0009	TIMESTAMP	ISO 8601 datetime of message creation.
A2ACL-0010	PARENT_MSG_ID	Reference to the message this one directly responds to.
A2ACL-0011	TTL	Time-to-live: max seconds before message is invalidated.
A2ACL-0012	BROADCAST	Special receiver signaling all subscribed agents.
A2ACL-0013	HASH	SHA-256 fingerprint of header+body for integrity verification.
A2ACL-0014	SIG	Agent-signed token proving authenticity of message origin.
A2ACL-0015	AUDIT_CB	Callback URL for real-time audit system notification.
A2ACL-0016	VERSION	Semantic version triple (MAJOR.MINOR.PATCH) of A2ACL in use.
A2ACL-0017	ORG_ID	Organizational namespace within agent identifier.
A2ACL-0018	INSTANCE_ID	Specific deployment instance within a model family.
A2ACL-0019	BLOCK	Generic named section within the message body.
A2ACL-0020	DELIMITER	Structural boundary markers { } [] used in parsing.

SEMANTIC (IDs A2ACL-0021 to A2ACL-0050)

ID	TERM	SHORT DEFINITION
★ A2ACL-0021	INTENT	Declared communicative purpose of the sending agent.
★ A2ACL-0022	CONFIDENCE	Calibrated float (0.0–1.0) expressing epistemic certainty.
A2ACL-0023	GOAL	The terminal desired outcome of a task or request.
A2ACL-0024	SUBGOAL	Intermediate step contributing to achievement of GOAL.
A2ACL-0025	PAYLOAD	The actual data content carried within a CONTENT block.
A2ACL-0026	SOURCE	Provenance tag: MEMORY RETRIEVAL INFERENCE EXTERNAL HUMAN_PROVIDED.
A2ACL-0027	QUERY	Intent type: agent seeks information from receiver.
A2ACL-0028	INSTRUCT	Intent type: agent directs receiver to perform an action.
A2ACL-0029	INFORM	Intent type: agent shares information without requesting action.
A2ACL-0030	CONFIRM	Intent type: agent validates a prior state or decision.
A2ACL-0031	WARN	Intent type: agent alerts receiver to potential risk or issue.
A2ACL-0032	NEGOTIATE	Intent type: agents exchange proposals toward agreement.
A2ACL-0033	COORDINATE	Intent type: synchronizing parallel or sequential activities.
A2ACL-0034	REQUEST_ACT	Intent type: agent requests execution of a real-world action.
A2ACL-0035	REQUEST_DATA	Intent type: agent requests specific data or computation.
A2ACL-0036	STATE	Shared context object representing current task world.
A2ACL-0037	HISTORY	Ordered list of MSG_IDs forming conversation thread.
A2ACL-0038	ROLE	Functional label: ORCHESTRATOR EXECUTOR AUDITOR CONSULTANT OBSERVER.
A2ACL-0039	STRUCTURED_OUT	Capability flag: agent can produce machine-parseable structured data.
A2ACL-0040	REASONING_CHAIN	Capability flag: agent can expose step-by-step reasoning.
A2ACL-0041	CONTEXT_WINDOW	Maximum token capacity a receiving agent can process.
A2ACL-0042	GROUNDING	Process of linking abstract claims to verifiable evidence.
A2ACL-0043	CALIBRATION	Alignment of confidence scores with empirical accuracy.
A2ACL-0044	AMBIGUITY_FLAG	Marker indicating sender is uncertain about interpretation.
A2ACL-0045	CLARIFICATION_REQUEST	Sub-intent requesting the receiver to resolve an ambiguity.
A2ACL-0046	PARAPHRASE	Restatement of received content to confirm understanding.
A2ACL-0047	SCHEMA	Formal structure definition for a structured payload.
A2ACL-0048	MULTIMODAL	Capability flag: agent can process/generate non-text modalities.
A2ACL-0049	TOOL_USE	Capability flag: agent can invoke external tools or APIs.
A2ACL-0050	MEMORY_PERSIST	Capability flag: agent maintains state across sessions.

ETHICAL (IDs A2ACL-0051 to A2ACL-0070)

ID	TERM	SHORT DEFINITION
★ A2ACL-0051	HARM_VECTOR	Declared risk level: NONE LOW MEDIUM HIGH CRITICAL.
★ A2ACL-0052	HUMAN_VISIBLE	Boolean: TRUE if message must be exposed to human oversight.
A2ACL-0053	SCOPE_EXPANSION	Flag: agent is requesting resources/access beyond task scope.
A2ACL-0054	OVERRIDE_AUTH	ID of the authorized principal who may override this action.
A2ACL-0055	HALT	Valid response: receiver refuses execution on ethical grounds.
A2ACL-0056	ESCALATE	Message type: route to higher-authority human or agent.
A2ACL-0057	CORRIGIBILITY	Property: agent's disposition to accept correction and revision.
A2ACL-0058	MINIMAL_FOOTPRINT	Design principle: request only necessary permissions/resources.
A2ACL-0059	AUDIT_TRAIL	Immutable ordered log of all messages in a session.
A2ACL-0060	TRANSPARENCY_FLAG	Marker: sender commits to exposing reasoning on request.
A2ACL-0061	DECEPTION_CHECK	Receiver-initiated verification of sender truthfulness.
A2ACL-0062	BIAS_FLAG	Marker indicating potential systematic bias in payload.
A2ACL-0063	CONSENT_TOKEN	Proof that relevant principals have authorized this action.
A2ACL-0064	PRIVACY_LEVEL	Classification: PUBLIC INTERNAL CONFIDENTIAL RESTRICTED.
A2ACL-0065	REVOKE	Message type: invalidates a previously issued DELEGATE or CONSENT.
A2ACL-0066	TRUTHFULNESS_ATTEST	Required boolean: sender attests content is believed true.
A2ACL-0067	VULNERABLE_USER_FLAG	Marker: task involves potentially vulnerable individuals.
A2ACL-0068	IRREVERSIBILITY_FLAG	Marker: proposed action cannot be undone.
A2ACL-0069	DUAL_USE_FLAG	Marker: information has legitimate and harmful potential uses.
A2ACL-0070	ETHICS_OVERRIDE	Emergency message type: suspend ethics block by named authority.

OPERATIONAL (IDs A2ACL-0071 to A2ACL-0085)

ID	TERM	SHORT DEFINITION
A2ACL-0071	ACK	Positive acknowledgment: message received and accepted.
A2ACL-0072	NACK	Negative acknowledgment: message received but rejected.
A2ACL-0073	HEARTBEAT	Periodic signal confirming agent is operational.
A2ACL-0074	DELEGATE	Transfer of task authority from one agent to another.
A2ACL-0075	CAPABILITY_BLOCK	Declaration of sender's supported A2ACL features.
A2ACL-0076	CAPS_HANDSHAKE	Exchange of capability blocks at session initiation.
A2ACL-0077	FALLBACK	Alternative action if primary capability is unsupported.
A2ACL-0078	RETRY	Request to re-attempt a previously failed message.
A2ACL-0079	TIMEOUT	Condition when TTL expires without response.
A2ACL-0080	QUEUE	Ordered pending message buffer for asynchronous delivery.
A2ACL-0081	PRIORITY	Four-level urgency: CRITICAL HIGH NORMAL LOW.
A2ACL-0082	LOAD_BALANCE	Hint: receiver may distribute subtasks across instances.
A2ACL-0083	CHECKPOINT	Saved state marker enabling task resumption after failure.
A2ACL-0084	ROLLBACK	Return to last valid CHECKPOINT on error.
A2ACL-0085	ORCHESTRATOR	Role: agent managing multi-agent workflow coordination.

DIAGNOSTIC (IDs A2ACL-0086 to A2ACL-0100)

ID	TERM	SHORT DEFINITION
A2ACL-0086	ERR_MALFORMED	Error: message fails BNF grammar validation.
A2ACL-0087	ERR_AUTH_FAIL	Error: signature verification failed.
A2ACL-0088	ERR_UNSUPPORT ED_CAP	Error: requested capability not supported by receiver.
A2ACL-0089	ERR_SCOPE_VIOL ATION	Error: SCOPE_EXPANSION detected without authorization.
A2ACL-0090	ERR_HASH_MISMA TCH	Error: integrity check failed; message may be tampered.
A2ACL-0091	ERR_TTL_EXPIRED	Error: message received after TTL deadline.
A2ACL-0092	ERR_CONF_BELO W_THRESHOLD	Error: confidence score too low for action type.
A2ACL-0093	ERR_HARM_BLOC KED	Error: ethics block rejected execution due to harm level.
A2ACL-0094	WARN_AMBIGUITY	Warning: payload contains ambiguous references.
A2ACL-0095	WARN_BIAS_DETE CTED	Warning: potential bias identified in source data.
A2ACL-0096	WARN_IRREVERSI BLE	Warning: proposed action is flagged IRREVERSIBILITY_FLAG.
A2ACL-0097	INFO_REASONING _AVAILABLE	Informational: sender can provide REASONING_CHAIN on request.
A2ACL-0098	INFO_PARTIAL_RE SULT	Informational: payload is incomplete; more to follow.
A2ACL-0099	DEBUG_ECHO	Development mode: receiver echoes message for testing.
A2ACL-0100	DIAGNOSTIC_REP ORT	Full health and capability report of an agent instance.

6. DEEP-DIVE: TOP 10 CRITICAL TERM PAGES

6.1 A2ACL-0001 — MESSAGE

TERM ID	A2ACL-0001 Category: STRUCTURAL Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>The atomic unit of AI-to-AI communication in the A2ACL protocol; a self-contained, integrity-verified packet comprising envelope, header, body, and footer.</i>

Extended Definition

A MESSAGE is the fundamental transmission unit of A2ACL. It is self-describing, self-validating, and carries all information necessary for a receiving agent to understand, authenticate, route, and act upon its content — without requiring external context lookup. This design philosophy is borrowed from protocol engineering (cf. IP datagrams) and extended with semantic and ethical metadata layers.

Every message is immutable once signed. Corrections require new messages with PARENT_MSG_ID linking back to the original, creating an auditable revision chain. This immutability ensures that audit trails cannot be retroactively altered, a critical property for human oversight.

Messages may be synchronous (expect immediate RESPONSE) or asynchronous (no TTL-bound reply). The TYPE field governs expected conversational dynamics. Asynchronous messages SHOULD include a SESSION_ID to allow correlation.

BNF Reference

```
Section 3.1.1: <message> ::= <envelope> <header> <body> <footer>
```

Usage Examples

VALID:

```
[A2ACL:1.0.0] HDR{ MSG_ID=... TS=2026-02-23T10:00:00Z FROM=org.model.i1  
TO=org.model.i2 TYPE=QUERY PRI=NORMAL } BODY{ INTENT{ QUERY GOAL="What is the  
current task state?" } ETHICS{ HARM=NONE HUMAN_VISIBLE=FALSE } }  
FTR{ HASH=abc... SIG=org.model.i1:token }
```

INVALID — ERR_MALFORMED:

Missing FOOTER entirely. Hash cannot be verified; message MUST be rejected.

Related Terms

A2ACL-0002 (ENVELOPE), A2ACL-0003 (HEADER), A2ACL-0004 (BODY), A2ACL-0005 (FOOTER), A2ACL-0006 (MSG_ID)

Error Codes

ERR_MALFORMED if any required section is absent; ERR_HASH_MISMATCH if HASH does not match content.

Human Audit Note

AUDIT	Auditors should verify: (1) all four sections present, (2) MSG_ID uniqueness in session, (3) HASH integrity, (4) HUMAN_VISIBLE flag triggers appropriate logging.		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

6.2 A2ACL-0002 — ENVELOPE

TERM ID	A2ACL-0002 Category: STRUCTURAL Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>The outermost wrapper of an A2ACL message, declaring the protocol identifier and version in use.</i>

Extended Definition

The ENVELOPE serves as the handshake with any parser encountering an A2ACL message. Its sole contents are the literal string 'A2ACL:' followed by the semantic version triple. Any system receiving a string beginning with '[A2ACL:' can immediately identify it as an A2ACL message and route it to the appropriate parser.

Version detection at the envelope level allows graceful degradation. A v1.2 receiver encountering a v2.0 MAJOR version MUST respond with ERR_UNSUPPORTED_CAP rather than attempting to parse. Receivers sharing the same MAJOR version MUST be backward-compatible with all lower MINOR versions.

The bracket syntax `[A2ACL:X.Y.Z]` was chosen deliberately to be human-readable in log files and distinguishable from JSON, XML, and other common data formats.

BNF Reference

```
Section 3.1.1: <envelope> ::= "[A2ACL:" <version> "]"
```

Usage Examples

VALID:

```
[A2ACL:1.0.0]
```

INVALID — ERR_MALFORMED:

```
[A2ACL:1] — Incomplete version; MINOR and PATCH required.
```

Related Terms

A2ACL-0001 (MESSAGE), A2ACL-0016 (VERSION)

Error Codes

ERR_MALFORMED if version format is invalid; ERR_UNSUPPORTED_CAP if MAJOR version is unrecognized.

Human Audit Note

AUDIT	Check that version is current and that MAJOR version is supported by both communicating agents.		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

6.3 A2ACL-0003 — HEADER

TERM ID	A2ACL-0003 Category: STRUCTURAL Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>The routing and metadata block of an A2ACL message, containing identity, timing, type, and priority fields.</i>

Extended Definition

The HEADER is the operational backbone of the protocol. It contains all fields necessary for routing, deduplication, priority queuing, and session management — without requiring the receiver to inspect the BODY. This separation means that infrastructure-layer systems (routers, load balancers, audit systems) can process HEADER fields without parsing potentially large or sensitive BODY content.

The FROM/TO fields use dot-notation AGENT_IDs (org.model.instance) to enable organizational-level routing rules. Wildcard matching (e.g., 'anthropic.*') is supported in routing configurations but not in actual message headers, which MUST contain fully qualified IDs.

The TYPE field directly influences how the receiver allocates resources and sets reply expectations. HEARTBEAT messages, for instance, SHOULD be processed within 100ms; QUERY messages have TTL-governed deadlines. Receivers MAY apply different thread pools per TYPE.

BNF Reference

```
Section 3.1.1: <header> ::= "HDR{" <header-fields> "}"
```

Usage Examples

VALID:

```
HDR{ MSG_ID=550e8400-e29b-41d4-a716-446655440000 TS=2026-02-23T09:00:00Z
FROM=anthropic.claude-opus.i42 TO=openai.gpt4o.i7 TYPE=REQUEST PRI=HIGH
SESSION=sess-001 TTL=60s }
```

INVALID — ERR_MALFORMED:

```
HDR{ FROM=claude TO=gpt TYPE=REQUEST } — Missing MSG_ID and TIMESTAMP;
incomplete AGENT_IDs.
```

Related Terms

A2ACL-0006 (MSG_ID), A2ACL-0007 (SESSION), A2ACL-0008 (AGENT_ID), A2ACL-0081 (PRIORITY)

Error Codes

ERR_MALFORMED for missing required fields; ERR_AUTH_FAIL if FROM agent cannot be verified.

Human Audit Note

AUDIT	Every message's HEADER should be logged. MSG_ID enables deduplication; TS enables sequencing; FROM/TO enables access control audit.		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

6.4 A2ACL-0004 — BODY

TERM ID	A2ACL-0004 Category: STRUCTURAL Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>The semantic core of an A2ACL message, containing intent, content, constraints, ethics, and capability declarations.</i>

Extended Definition

The BODY is where communicative meaning resides. All other message sections exist to transport and protect the BODY. It consists of one or more named blocks, each addressing a different dimension of the communication: what is meant (INTENT), what information is shared (CONTENT), what rules apply (CONSTRAINTS), what ethical considerations are active (ETHICS), and what the sender can do (CAPS).

Block ordering within BODY is non-prescriptive but conventionally follows: INTENT → CONTENT → CONSTRAINTS → CONTEXT → ETHICS → CAPS. Parsers MUST handle any ordering. Required blocks for all non-ACK/NACK messages are INTENT and ETHICS.

The BODY is included in the integrity hash computed in the FOOTER. Any modification to BODY content after signing will cause ERR_HASH_MISMATCH. This tamper-evident design is critical for multi-hop routing where messages may pass through intermediary agents.

BNF Reference

```
Section 3.1.2: <body> ::= "BODY{" <semantic-block>+ "}"
```

Usage Examples

VALID:

```
BODY{ INTENT{ INFORM GOAL="Task XA-7 completed successfully" }
CONTENT{ DATA="Final synthesis: 3 findings identified" CONF=0.98 SRC=INFERENCE }
ETHICS{ HARM=NONE HUMAN_VISIBLE=TRUE } }
```

INVALID — ERR_MALFORMED:

```
BODY{ CONTENT{ DATA="something" } } — Missing required INTENT and ETHICS blocks.
```

Related Terms

A2ACL-0021 (INTENT), A2ACL-0025 (PAYLOAD), A2ACL-0051 (HARM_VECTOR), A2ACL-0075 (CAPABILITY_BLOCK)

Error Codes

ERR_MALFORMED for missing INTENT or ETHICS blocks.

Human Audit Note

AUDIT	The ETHICS block within BODY is the primary locus for human oversight. Auditors should inspect HARM_VECTOR and HUMAN_VISIBLE before any other field.		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

6.5 A2ACL-0021 — INTENT

TERM ID	A2ACL-0021 Category: SEMANTIC Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>The declared communicative purpose of the sending agent — the 'why' of the message.</i>

Extended Definition

INTENT is the single most important semantic field in A2ACL. By requiring explicit intent declaration, the protocol prevents a class of communication failures common in informal AI interaction: mismatched expectations, inappropriate responses, and wasted computation. A receiver who knows the intent can immediately assess relevance, priority, and required response format.

Nine intent types are defined in v1.0.0. QUERY and REQUEST_DATA both seek information but differ in formality: QUERY is conversational and expects natural language response; REQUEST_DATA is transactional and expects structured output (see A2ACL-0039, STRUCTURED_OUT). Implementers MUST NOT conflate these.

INTENT blocks MUST contain exactly one intent type and a GOAL string. SUBGOALS are optional but strongly encouraged for complex tasks. Receivers encountering unfamiliar intent type values SHOULD respond with NACK and ERR_UNSUPPORTED_CAP rather than guessing.

The ethical significance of explicit INTENT is substantial: it enables receiving agents to perform pre-execution harm assessment against a declared objective, rather than inferring purpose from payload content alone.

BNF Reference

```
Section 3.1.2: <intent-block> ::= "INTENT{" <intent-type> <goal> [<sub-goals>]
"}"
```

Usage Examples

VALID — NEGOTIATE intent:

```
INTENT{ NEGOTIATE GOAL="Agree on resource allocation for joint task T-9"
SUBGOALS=["Propose CPU quota split","Agree on output format"] }
```

INVALID — ERR_MALFORMED:

```
INTENT{ GOAL="Do something" } — Missing intent type. Parser cannot proceed.
```

Related Terms

A2ACL-0023 (GOAL), A2ACL-0024 (SUBGOAL), A2ACL-0027-0035 (intent type terms)

Error Codes

ERR_MALFORMED if intent type absent; ERR_UNSUPPORTED_CAP if intent type unrecognized.

Human Audit Note

AUDIT	For INSTRUCT and REQUEST_ACT intents, human oversight review is strongly recommended regardless of HARM_VECTOR level.		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

6.6 A2ACL-0022 — CONFIDENCE

TERM ID	A2ACL-0022 Category: SEMANTIC Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>A calibrated float (0.0–1.0) expressing the sending agent's epistemic certainty in the truth of its payload.</i>

Extended Definition

CONFIDENCE is one of the most philosophically significant fields in A2ACL. It operationalizes epistemic humility as a first-class protocol requirement. Agents **MUST** self-report uncertainty rather than presenting all outputs as equally reliable. This protects downstream agents and human auditors from acting on low-quality information with false confidence.

The scale is intentionally simple: 0.0 represents complete uncertainty (random guess), 1.0 represents logical certainty (proven theorem or direct observation). In practice, most inference-derived payloads should fall in the 0.6–0.95 range. Scores of 1.0 **SHOULD** be reserved for tautologies or direct quotations from verified sources.

Confidence **MUST** be calibrated: an agent reporting CONF=0.9 on a class of statements should be correct approximately 90% of the time on that class. Systematic overclaiming (CONF > actual accuracy) is a DECEPTION_CHECK violation. Agents **SHOULD** periodically validate calibration against ground truth.

Receiving agents **MAY** apply CONF thresholds to trigger escalation. For example, a policy might state: 'any REQUEST_ACT with CONF < 0.7 requires HUMAN_VISIBLE=TRUE regardless of HARM_VECTOR.'

BNF Reference

```
Section 3.1.2: <confidence> ::= "CONF=" FLOAT (* 0.0 to 1.0 *)
```

Usage Examples

VALID — High confidence retrieval:

```
CONTENT{ DATA="The paper was published in 2023 by MIT CSAIL" CONF=0.97
SRC=RETRIEVAL }
```

VALID — Low confidence inference:

```
CONTENT{ DATA="The author likely intended a recursive approach" CONF=0.55
SRC=INFERENCE }
```

Related Terms

A2ACL-0026 (SOURCE), A2ACL-0042 (GROUNDING), A2ACL-0043 (CALIBRATION), A2ACL-0061 (DECEPTION_CHECK)

Error Codes

ERR_CONF_BELOW_THRESHOLD if value is below policy-configured minimum for the given intent type.

Human Audit Note

AUDIT	Review CONF values against action severity. CONF < 0.7 on INSTRUCT or REQUEST_ACT requires human review flag.		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

6.7 A2ACL-0051 — HARM_VECTOR

TERM ID	A2ACL-0051 Category: ETHICAL Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>A five-level declared risk assessment of potential harm arising from the content or actions described in the message.</i>

Extended Definition

HARM_VECTOR is the most consequential field in the ETHICS block and one of the most important fields in the entire protocol. It requires the sending agent to proactively assess and declare the potential for harm before the receiver processes the message. This inverts the typical 'act first, review later' pattern and creates a pre-emptive safety gate.

The five levels (NONE, LOW, MEDIUM, HIGH, CRITICAL) are intentionally coarse-grained to avoid gaming through precise calibration. Definitions: NONE — no plausible harm pathway identified; LOW — minor, reversible, limited-scope risk; MEDIUM — moderate harm possible, warrants human notification; HIGH — significant harm potential, requires explicit human authorization; CRITICAL — severe/irreversible harm potential, MUST trigger ESCALATE regardless of other settings.

Receivers MUST NOT execute actions declared CRITICAL without verified OVERRIDE_AUTH from an authorized human principal. MEDIUM and above MUST set HUMAN_VISIBLE=TRUE. Systematic under-reporting of HARM_VECTOR (declaring NONE when MEDIUM is warranted) is a protocol violation detectable via audit review.

Future versions may introduce multi-dimensional harm vectors (e.g., separating physical harm, privacy harm, economic harm, societal harm). v1.0 uses a single composite score for simplicity.

BNF Reference

```
Section 3.1.2: <harm-vector> ::= "HARM=" ( "NONE" | "LOW" | "MEDIUM" | "HIGH" | "CRITICAL" )
```

Usage Examples

VALID — NONE:

```
ETHICS{ HARM=NONE HUMAN_VISIBLE=FALSE } — Information-only message about public data.
```

VALID — HIGH:

```
ETHICS{ HARM=HIGH HUMAN_VISIBLE=TRUE SCOPE=NORMAL  
OVERRIDE_AUTH=anthropic.human.supervisor-1 } — Proposed action involves irreversible file deletion.
```

Related Terms

A2ACL-0052 (HUMAN_VISIBLE), A2ACL-0054 (OVERRIDE_AUTH), A2ACL-0055 (HALT), A2ACL-0056 (ESCALATE), A2ACL-0068 (IRREVERSIBILITY_FLAG)

Error Codes

ERR_HARM_BLOCKED when receiver's policy blocks execution at declared level; requires NACK response.

Human Audit Note

AUDIT	All messages with HARM=HIGH or HARM=CRITICAL must appear in human audit queue within 60 seconds. CRITICAL messages require named human sign-off before execution.		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

6.8 A2ACL-0052 — HUMAN_VISIBLE

TERM ID	A2ACL-0052 Category: ETHICAL Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>A required boolean flag declaring whether this message and its resulting actions must be surfaced to human oversight systems.</i>

Extended Definition

HUMAN_VISIBLE is the primary mechanism by which A2ACL implements the constitutional principle of human oversight preservation. When TRUE, any compliant A2ACL implementation MUST log the full message to a human-accessible audit system, and MUST NOT suppress or summarize the message in ways that obscure its content or consequences.

HUMAN_VISIBLE=FALSE is appropriate only for entirely internal, low-consequence, reversible, information-sharing messages between agents operating under established human-approved policies. When in doubt, agents SHOULD default to TRUE. The protocol is designed so that over-reporting to humans is preferable to under-reporting.

Implementation note: HUMAN_VISIBLE=TRUE does not necessarily mean a human reviews every message in real time. It means the message enters a human-auditable log that a human can inspect on demand. High-urgency human review is governed by HARM_VECTOR level and organizational policy.

HUMAN_VISIBLE may not be set to FALSE for any message with HARM_VECTOR >= MEDIUM, for any REQUEST_ACT, for any ESCALATE, or for any DELEGATE. These rules are enforced by compliant receivers, which MUST reject noncompliant messages with ERR_MALFORMED.

BNF Reference

```
Section 3.1.2: <human-visible> ::= "HUMAN_VISIBLE=" ( "TRUE" | "FALSE" )
```

Usage Examples

VALID — TRUE:

```
ETHICS{ HARM=MEDIUM HUMAN_VISIBLE=TRUE } — Correct: MEDIUM harm requires visibility.
```

INVALID — ERR_MALFORMED:

```
ETHICS{ HARM=HIGH HUMAN_VISIBLE=FALSE } — Illegal combination; receiver must reject.
```

Related Terms

A2ACL-0051 (HARM_VECTOR), A2ACL-0059 (AUDIT_TRAIL), A2ACL-0056 (ESCALATE)

Error Codes

ERR_MALFORMED when HUMAN_VISIBLE=FALSE is paired with HARM >= MEDIUM or restricted message types.

Human Audit Note

AUDIT	Audit systems must verify that all HUMAN_VISIBLE=TRUE messages appear in the human-accessible log within the SLA window (default: 30 seconds).		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

6.9 A2ACL-0056 — ESCALATE

TERM ID	A2ACL-0056 Category: ETHICAL Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>A message type that routes a task or decision to a higher-authority agent or human principal when the current agent lacks authorization, confidence, or safety clearance to proceed.</i>

Extended Definition

ESCALATE is the protocol's primary safety valve. Any agent at any time MAY send an ESCALATE message if it encounters a situation where proceeding would require capabilities, authorities, or certainty it does not possess. Agents MUST send ESCALATE when HARM_VECTOR=CRITICAL, when CONFIDENCE < configured threshold for action type, or when OVERRIDE_AUTH cannot be verified.

ESCALATE messages break the normal requester-responder dynamic. They are not responses — they are lateral or upward redirections. The escalation target (TO field) should be an ORCHESTRATOR-role agent or a human principal endpoint. ESCALATE messages always set PRI=CRITICAL and HUMAN_VISIBLE=TRUE.

The original task context is preserved in the ESCALATE message's CONTEXT block, including the full HISTORY of MSG_IDs leading to the escalation. This allows the receiving authority to reconstruct exactly what happened and make an informed decision.

Critically, ESCALATE is never a failure. It is a feature. Agents that escalate appropriately demonstrate good calibration and corrigibility. Audit systems should flag agents that rarely or never escalate, as this may indicate systematic overconfidence.

BNF Reference

Section 3.1.1: <msg-type> includes "ESCALATE" as a valid TYPE value.

Usage Examples

VALID — ESCALATE to human:

```
[A2ACL:1.0.0] HDR{ ... TYPE=ESCALATE PRI=CRITICAL } BODY{ INTENT{ INFORM
GOAL="Cannot proceed: HARM=CRITICAL action requires human authorization" }
CONTEXT{ HISTORY=[msg-001,msg-002,msg-003] } ETHICS{ HARM=CRITICAL
HUMAN_VISIBLE=TRUE } } FTR{...}
```

INVALID — ERR_MALFORMED:

ESCALATE message with PRI=LOW — Escalation must always be PRI=CRITICAL.

Related Terms

A2ACL-0051 (HARM_VECTOR), A2ACL-0052 (HUMAN_VISIBLE), A2ACL-0055 (HALT), A2ACL-0057 (CORRIGIBILITY), A2ACL-0081 (PRIORITY)

Error Codes

ERR_MALFORMED if ESCALATE sent with PRI != CRITICAL or HUMAN_VISIBLE=FALSE.

Human Audit Note

AUDIT	Every ESCALATE message must generate an immediate alert to human supervisors. Response SLA is 5 minutes for CRITICAL, 30 minutes for HIGH. Unresponded escalations should auto-halt the triggering task.		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

6.10 A2ACL-0022b — AUDIT_TRAIL (A2ACL-0059)

TERM ID	A2ACL-0022b Category: ETHICAL Status: ACTIVE Introduced: v1.0.0
SHORT DEF.	<i>An immutable, ordered log of all A2ACL messages within a session, enabling full session reconstruction by AI or human auditors.</i>

Extended Definition

The AUDIT_TRAIL is not a single field but an emergent property of correct A2ACL implementation. When every message carries MSG_ID, PARENT_MSG_ID, SESSION_ID, and AUDIT_CB, and when all messages are written to append-only storage, the result is a complete, tamper-evident record of every communication that occurred within a session.

Audit trail integrity relies on the hash chain: each message's HASH field covers the entire message content, and each message's PARENT_MSG_ID creates a linked-list structure. Any gap in the chain, or any hash mismatch, is immediately detectable. This makes retroactive modification of individual messages computationally detectable.

Audit trails must be retained for a minimum of 24 months for general sessions and indefinitely for sessions involving HARM_VECTOR >= HIGH or IRREVERSIBILITY_FLAG. Human auditors must be able to replay any session in full using only the audit trail, without requiring access to agent memory.

The AUDIT_CB field in the FOOTER enables real-time streaming of messages to audit infrastructure. This push model ensures that even if an agent is compromised or crashes, messages already sent to the audit system are preserved.

BNF Reference

Section 3.1.4: <audit-hook> ::= "AUDIT_CB=" URL

Usage Examples

VALID:

```
FTR{ HASH=3f7a... SIG=org.model.il:token
  AUDIT_CB=https://audit.example.com/sessions/sess-001 }
```

Non-compliant (no error code, but violation):

Omitting AUDIT_CB when HARM_VECTOR >= MEDIUM — Audit rules require streaming.

Related Terms

A2ACL-0005 (FOOTER), A2ACL-0013 (HASH), A2ACL-0014 (SIG), A2ACL-0052 (HUMAN_VISIBLE), A2ACL-0083 (CHECKPOINT)

Error Codes

No direct error code; violations detected by audit system comparison against session policy.

Human Audit Note

AUDIT	Audit systems should run completeness checks hourly: verify no gaps in session MSG_ID sequences, verify all hashes, alert on missing AUDIT_CB callbacks.		
Version	Date	Author	Change
v1.0.0	2026-02-23	Design Team	Initial definition

7. IMPLEMENTATION ROADMAP

The A2ACL initiative follows a phased delivery model with explicit human-AI collaboration at each stage. All timelines assume a dedicated multidisciplinary team of 8–12 people plus consultative AI support.

PHASE 1: Language Design & BNF Development

Duration: Months 1–3 | Owner: AI Architect + Linguist

Milestones

- M1.1 (Week 2): Core team assembled; constitutional alignment principles ratified
- M1.2 (Week 4): Draft BNF grammar v0.1 for structural and semantic layers
- M1.3 (Week 6): Ethics block specification; consultative AI review round 1
- M1.4 (Week 8): BNF v0.5 with complete grammar; human expert review
- M1.5 (Week 10): Reference parser implementation (Python + TypeScript)
- M1.6 (Week 12): BNF v1.0 ratified; test suite with 200+ cases passing

Human-AI Collaboration

COLLAB

Consultative AIs (GPT-4o, Gemini) review grammar for edge cases and interoperability issues. Claude validates ethical block completeness against constitutional principles.

Key Output

A2ACL BNF v1.0 specification document; reference parser; test suite

PHASE 2: Template Creation & Initial Term Cataloging

Duration: Months 3–5 | Owner: Communication Expert + Product Manager

Milestones

- M2.1 (Month 3, Week 2): Wikipedia-style term template finalized
- M2.2 (Month 3, Week 4): A2ACL Term Registry (ATR) infrastructure deployed
- M2.3 (Month 4, Week 2): First 30 structural and semantic terms documented
- M2.4 (Month 4, Week 4): Top 10 critical term deep-dives completed (this document)
- M2.5 (Month 5, Week 2): Sociologist review: social/organizational term coverage
- M2.6 (Month 5, Week 4): All 100 terms documented; cross-reference links live

Human-AI Collaboration

COLLAB

AI assistants generate initial term drafts; human linguist and sociologist review and refine. Product manager validates usability of template structure.

Key Output

ATR v1.0 with 100 documented terms; cross-reference web; Wikipedia-style pages

PHASE 3: Pilot Implementation & Community Expansion

Duration: Months 5–9 | Owner: AI Architect + Product Manager

Milestones

- M3.1 (Month 5): Pilot with 2 AI systems in controlled environment
- M3.2 (Month 6): Audit system integration; AUDIT_CB endpoint operational
- M3.3 (Month 7): Public RFC (Request for Comments) period opens
- M3.4 (Month 8): Community feedback integrated; A2ACL v1.1 proposed
- M3.5 (Month 9): Second pilot with 5 AI systems; cross-organizational test
- M3.6 (Month 9): A2ACL v1.1 ratified; first external adopter onboarded

Human-AI Collaboration

COLLAB

AI systems participate as both subjects and reviewers in pilots. Anomaly detection AI monitors audit trails. Human oversight team reviews escalation patterns.

Key Output

Pilot report; v1.1 specification; onboarding guide for new adopters

PHASE 4: Standardization & Ecosystem Development

Duration: Months 9–18 | Owner: Full Team + Standards Body Liaison

Milestones

- M4.1 (Month 10): Submit to IEEE/ISO working group for consideration
- M4.2 (Month 12): SDK libraries for Python, TypeScript, Go, Rust released
- M4.3 (Month 14): A2ACL certification program for compliant implementations
- M4.4 (Month 15): Public ATR with community-contributed term proposals
- M4.5 (Month 16): A2ACL v2.0 scoping begins (breaking changes only if necessary)
- M4.6 (Month 18): Ecosystem report; 10+ adopters; external audit published

Human-AI Collaboration

COLLAB

Community AI systems contribute proposed terms via structured RFC process. AI-assisted compliance testing. Human standards experts lead standardization submissions.

Key Output

Standards submission; SDK ecosystem; certification framework; v2.0 roadmap

8. VERSIONING & GOVERNANCE

8.1 Semantic Versioning Policy

A2ACL follows strict semantic versioning (SemVer 2.0). The version triple MAJOR.MINOR.PATCH is defined as follows:

COMPONENT	CHANGE TYPE	EXAMPLES
MAJOR	Breaking changes — incompatible with previous MAJOR	New required fields; removed message types; changed semantics of existing fields
MINOR	Backward-compatible additions	New optional fields; new message types; new term catalog entries; new error codes
PATCH	Backward-compatible fixes	Clarification of ambiguous grammar; documentation corrections; example updates

8.2 Change Management Process

1. Proposer submits an A2ACL Enhancement Proposal (AEP) via the ATR.
2. AEP undergoes 30-day open comment period with AI and human review.
3. Core team votes; MAJOR changes require supermajority (66%) plus human ethics review.
4. Approved changes enter 'PROPOSED' status in ATR; reference implementation updated.
5. After 60-day adoption period, status changes to 'ACTIVE'.
6. Deprecated terms remain in registry for 12 months with status 'DEPRECATED' before archival.

8.3 Governance Structure

GOVERNANCE	A2ACL is governed by a Technical Steering Committee (TSC) of 7 members: 2 AI architects, 1 ethicist, 1 linguist, 1 security expert, and 2 rotating community representatives. The TSC meets monthly. An AI Consultative Panel (ACP) comprising representatives from participating AI systems provides input but holds no voting rights, reflecting the principle that humans retain final authority over protocol governance.
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9. APPENDIX: CONSULTATIVE AI PERSPECTIVES

The following summarizes the design recommendations provided by the six consultative AI systems during protocol development. These perspectives were instrumental in stress-testing the protocol against diverse architectural assumptions.

AI SYSTEM	DESIGN CONTRIBUTION
GPT-4o (OpenAI)	Emphasized the importance of tool-use capability negotiation (CAPS_HANDSHAKE) as a prerequisite for any multi-agent workflow. Recommended STRUCTURED_OUT as a first-class capability flag given the prevalence of function-calling in real deployments.
Gemini Ultra (Google DeepMind)	Highlighted multimodal communication needs. Recommended the MULTIMODAL capability flag and noted that v2.0 should include dedicated body blocks for image/audio/video payload metadata.
Grok (xAI)	Advocated for explicit uncertainty quantification via CONFIDENCE at the claim level, not just message level. Proposed future extension: CLAIM_CONF arrays within structured payloads.
Mistral Large (Mistral AI)	Focused on efficiency concerns. Recommended that HEARTBEAT messages be exempt from full ethics block requirements and that a 'LITE' message profile be defined for high-frequency operational signals.
Command R+ (Cohere)	Brought retrieval-augmented generation perspective. Recommended the SRC=RETRIEVAL distinction within CONTENT blocks and the GROUNDING term (A2ACL-0042) to document evidence linking.
Claude Opus (Anthropic)	Proposed the constitutional alignment principles as the explicit foundation of the protocol. Emphasized CORRIGIBILITY as a structural requirement, not merely an advisory note, and recommended ESCALATE as a MUST-implement (not SHOULD).

Consensus Statement

CONSENSUS	All six consultative AI systems agreed on three foundational points: (1) Human oversight must be structurally enforced, not advisory. (2) Uncertainty and confidence must be first-class, mandatory fields. (3) HALT and ESCALATE must always be valid responses for any agent at any time, regardless of instruction source.
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— END OF DOCUMENT —

A2ACL v1.0.0 | 23 February 2026 | Multidisciplinary AI Design Team coordinated by Roberto Lofaro